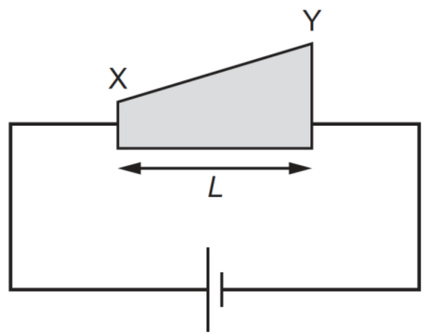
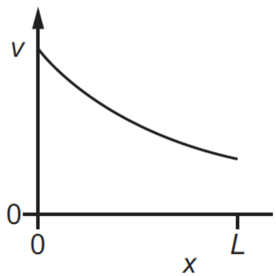
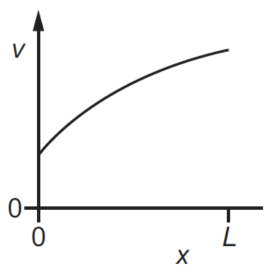
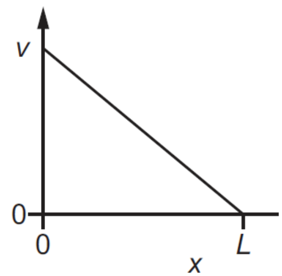
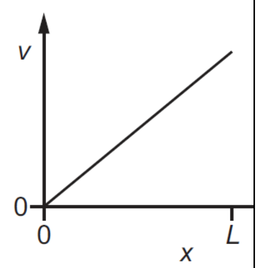
22 A wedge-shaped metal conductor of length L, varying width and uniform thickness is connected to a cell, as shown.



Which graph best shows how the average drift velocity v of electrons in the conductor varies with distance *x* from end X?

A	B	C	D
			

Answer: A

From $I = nAv_dq$

$I = nAv_dq$

$v_d = \frac{I}{nAq}$

The cross-sectional area of the metal conductor increases from X to Y but the current must remain constant. So the drift speed v_d of the electrons must decrease as they move from X

	to Y, eliminating graphs B and D. Graph C can also be rejected, as this graph implies that the electrons stop at Y.

23	A 9.0 V battery with negligible internal resistance is connected into a circuit, as shown